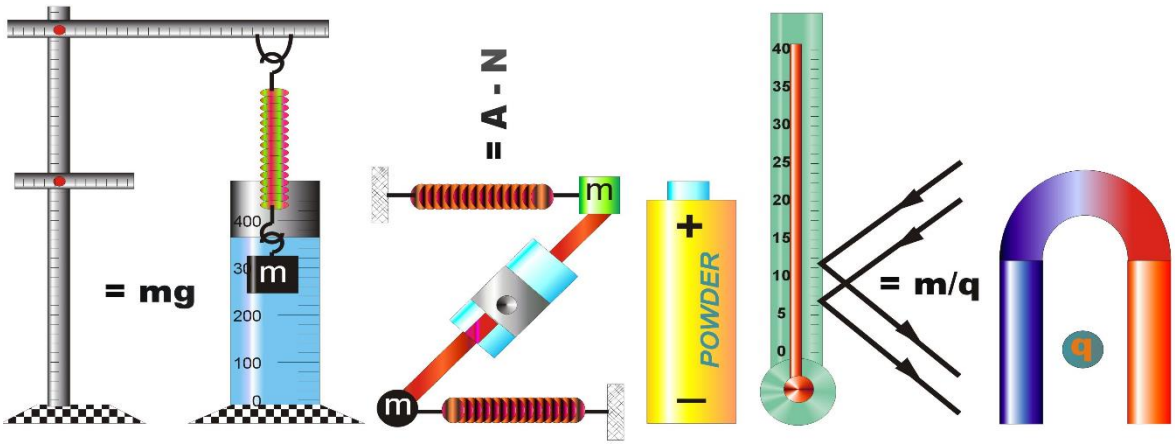
Toshpo‘latov Ma’ruf Abdumalikovich

O‘ZBEKISTON RESPUBLIKASI XALQ TA’LIMI VAZIRLIGI
SAMARQAND VILOYATI XALQ TA’LIMI BOSHQARMASI
BULUNG‘UR TUMANI XALQ TA’LIMI BO‘LIMI
67-UMUMIY O‘RTA TA’LIM MAKTAB

9-SINF



DARSLIKDAGI MASALALAR YECHIMLARI

BULUNG‘UR - 2019

1-MASHQ

①. Massasi 2,4 kg bo‘lgan ko‘mirda qancha uglerod atomi borligini hisoblang. Uglerod atomining massasini $2 \cdot 10^{-26}$ kg ga teng deb oling.

Berilgan:	Formulasi:	Hisoblash:
$m = 2{,}4\ kg$ $m_0 = 2 \cdot 10^{-26}\ kg$	$N = \frac{m}{m_0}$ $[N] = \frac{kg}{kg} = birliksiz$	$N = \frac{2{,}4}{2 \cdot 10^{-26}} = 1{,}2 \cdot 10^{26}\ ta$
Topish kerak: $N = ?$		**Javob:** $N = 1{,}2 \cdot 10^{26}\ ta$

②. Hajmi 0,2 mm^3 bo‘lgan moy suv ustida yoyilib, taxminan 0,8 m^2 yuzali yupqa parda hosil qildi. Moy molekulalari suv yuzasida bir qavat bo‘lib tekis yoyilgan deb hisoblab, moy molekulasining chiziqli o‘lchamini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$V = 0{,}2\ mm^3 = 2 \cdot 10^{-10}\ m^3$ $S = 0{,}8\ m^2$	$d = \frac{V}{S}$ $[d] = \frac{m^3}{m^2} = m$	$d = \frac{2 \cdot 10^{-10}}{0{,}8} = 2{,}5 \cdot 10^{-10}\ m$
Topish kerak: $d = ?$		**Javob:** $d = 2{,}5 \cdot 10^{-10}\ m$

③. Bitta suv molekulasining massasi $3 \cdot 10^{-26}$ kg. Hajmi 5 cm^3 bo‘lgan suvda qancha suv molekulasi bor?

Berilgan:	Formulasi:	Hisoblash:
$m_0 = 3 \cdot 10^{-26}\ kg$ $V = 5\ cm^3 = 5 \cdot 10^{-6}\ m^3$ $\rho = 1000\ kg/m^3$	$m = \rho \cdot V$ $N = \frac{m}{m_0} = \frac{\rho \cdot V}{m_0}$ $[N] = \frac{kg}{kg} = 1$	$N = \frac{1000 \cdot 5 \cdot 10^{-6}}{3 \cdot 10^{-26}} \approx 1{,}67 \cdot 10^{23}\ ta$
Topish kerak: $N = ?$		**Javob:** $N \approx 1{,}67 \cdot 10^{23}\ ta$

④*. Idishdagi suvda 10^{24} ta suv molekulasi bo‘lsa, suvning hajmi qanday? Suv molekulasining diametrini $3 \cdot 10^{-10}$ m ga teng deb oling.

Berilgan:	Formulasi:	Hisoblash:
$N = 10^{24}\ ta$ $d = 3 \cdot 10^{-10}\ m$ $m_0 = 2{,}7 \cdot 10^{-26}\ kg$ $\rho = 1000\ kg/m^3$	$N = \frac{m}{m_0} = \frac{\rho \cdot V}{m_0}$ $V = \frac{N \cdot m_0}{\rho}$ $[V] = \frac{kg}{kg/m^3} = m^3$	$V = \frac{10^{24} \cdot 2{,}7 \cdot 10^{-26}}{1000} = 2{,}7 \cdot 10^{-5}\ m^3$
Topish kerak: $V = ?$		**Javob:** $N = 2{,}7 \cdot 10^{-5}\ m^3 = 27\ cm^3$

⑤*. Moy molekulasining diametri taxminan $2{,}6 \cdot 10^{-10}$m bo‘lsa, 35 cm^3 moyda qancha moy molekulasi borligini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$d = 2{,}6 \cdot 10^{-10}\ m$ $V = 35\ cm^3 = 35 \cdot 10^{-6}\ m^3$	$N = \frac{V}{d^3}$	$N = \frac{35 \cdot 10^{-6}}{(2{,}6 \cdot 10^{-10})^3} \approx 2 \cdot 10^{24}\ ta$
Topish kerak: N =?	$[N] = \frac{m^3}{m^3} = 1$	**Javob:** $N \approx 2 \cdot 10^{24}\ ta$

⑥. Jadvalni to‘ldiring.

№	Modda	Kimyoviy formulasi	Nisbiy molekulyar massasi
1	Azot	N_2	28 u
2	Ozon	O_3	48 u
3	Osh tuzi	NaCl	58 u
4	Metan gazi	CH_4	16 u
5	Karbonat angdirid	CO_2	44 u



2-MASHQ

①. Massasi 270 g bo‘lgan suvdagi modda miqdorini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$m = 270\ g$ $M = 18\ g/mol$	$\nu = \frac{m}{M}$ $[\nu] = \frac{g}{g/mol} = mol$	$\nu = \frac{270}{18} = 15\ mol$
Topish kerak: ν =?		**Javob:** $\nu = 15\ mol$

②. Miqdori 8 mol bo‘lgan karbonat angidrid (CO_2) gazining massasi nimaga teng?

Berilgan:	Formulasi:	Hisoblash:
$\nu = 8\ mol$ $M_{CO_2} = 44\ g/mol$	$m = \nu \cdot M$	$m = 8 \cdot 44 = 352\ g$
Topish kerak: m =?	$[m] = mol \cdot \frac{g}{mol} = g$	**Javob:** $m = 352\ g$

③. Massasi 7 g bo‘lgan azot (N_2) tarkibidagi molekulalar sonini aniqlang.

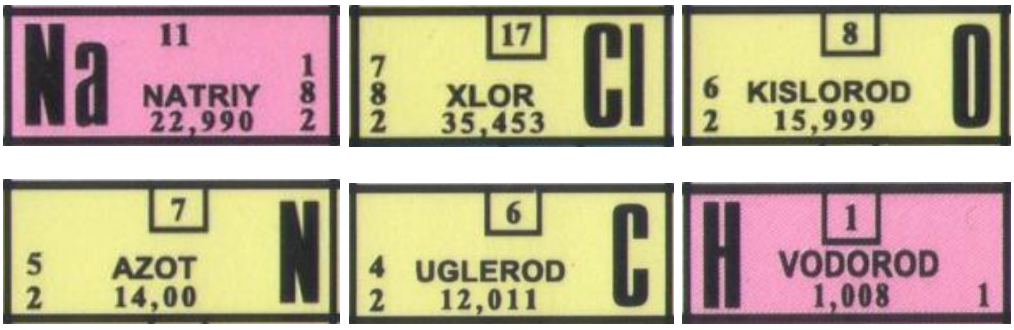
Berilgan:	Formulasi:	Hisoblash:
$m = 7\ g$ $M = 28\ g/mol$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$ *Topish kerak:* $N = ?$	$N = \frac{m}{M} \cdot N_A$ $[N] = \frac{g}{g/mol} \cdot \frac{1}{mol} = 1$	$N = \frac{7}{28} \cdot 6 \cdot 10^{23} = 1{,}5 \cdot 10^{23}\ ta$ **Javob:** $N = 1{,}5 \cdot 10^{23}\ ta$

④. Moddaning molyar massasi 36 g/molga teng bo‘lsa, shu modda bitta molekulasining massasini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$M = 36\frac{g}{mol} = 36 \cdot 10^{-3}\frac{kg}{mol}$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$ *Topish kerak:* $m_0 = ?$	$m_0 = \frac{M}{N_A}$ $[m_0] = \frac{\frac{kg}{mol}}{1/mol} = kg$	$m_0 = \frac{36 \cdot 10^{-3}}{6 \cdot 10^{23}} = 6 \cdot 10^{-26}\ kg$ **Javob:** $m_0 = 6 \cdot 10^{-26}\ kg$

⑤. Jadvalni to‘ldiring.

№	Moddaning turi	Kimyoviy formulasi	Nisbiy molekulyar massasi (g/mol)	Molekulaning massasi (g)
1	Osh tuzi	NaCl	58	$9{,}67 \cdot 10^{-26}$
2	Ozon	O_3	48	$8 \cdot 10^{-26}$
3	Azot	N_2	28	$4{,}67 \cdot 10^{-26}$
4	Metan gazi	CH_4	16	$2{,}67 \cdot 10^{-26}$



⑥. Bir dona gaz molekulasining massasi $7{,}33 \cdot 10^{-26}$ kg ga teng. Shu gazning molyar massasini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$m_0 = 7{,}33 \cdot 10^{-26}\ kg$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$ *Topish kerak:* $M = ?$	$M = m_0 \cdot N_A$ $[M] = kg \cdot \frac{1}{mol} = \frac{kg}{mol}$	$M = 7{,}33 \cdot 10^{-26} \cdot 6 \cdot 10^{23} =$ $= 0{,}044\ \frac{kg}{mol} = 44\ \frac{g}{mol}$ **Javob:** $M = 44\ g/mol$ (korbonat angdrid CO_2)

3-MASHQ

①. Massasi 81 g bo‘lgan buyum alyuminiydan yasalgan. Buyumda qancha alyuminiy atomi bor?

Berilgan:	Formulasi:	Hisoblash:
$m = 81\ g$ $M = 27\ g/mol$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$	$N = \frac{m}{M} \cdot N_A$ $[N] = \frac{g}{g/mol} \cdot \frac{1}{mol} = 1$	$N = \frac{81}{27} \cdot 6 \cdot 10^{23} = 1{,}8 \cdot 10^{24}\ ta$ **Javob:** $N = 1{,}8 \cdot 10^{24}\ ta$
Topish kerak: N =?		

②. $4 \cdot 10^{24}$ ta temir atomi bo‘lgan jismning massasi qanday?

Berilgan:	Formulasi:	Hisoblash:
$N = 4 \cdot 10^{24}$ $M = 56\ g/mol$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$	$m = \frac{N \cdot M}{N_A}$ $[m] = \frac{g/mol}{1/mol} = g$	$m = \frac{4 \cdot 10^{24} \cdot 56}{6 \cdot 10^{23}} = 373\ g$ **Javob:** $m = 373\ g$
Topish kerak: m =?		

③. Idishga solingan gazning massasi 5,5 kg ga teng. Idishda $7{,}5\ 10^{25}$ ta gaz molekulasi mavjud bo‘lsa, bu gaz turini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$m = 5{,}5\ kg = 5{,}5 \cdot 10^3\ g$ $N = 7{,}5 \cdot 10^{25}$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$	$M = \frac{m}{N} \cdot N_A$ $[M] = g \cdot \frac{1}{mol} = \frac{g}{mol}$	$M = \frac{5{,}5}{7{,}5 \cdot 10^{25}} \cdot 6 \cdot 10^{23} =$ $= 44\ g/mol$ **Javob:** $M = 44\ g/mol$ (korbonat angdrid CO_2)
Topish kerak: M =?		

④. Idishga massasi 72 g bo‘lgan suv solingan. Idishdagi suv molekulalarining konsentratsiyasini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$m = 72\ g = 72 \cdot 10^{-3}\ kg$ $\rho = 1000\ kg/m^3$ $M = 18 \cdot 10^{-3}\ kg/mol$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$	$N = \frac{m}{M} \cdot N_A$ $V = \frac{m}{\rho}$ $n = \frac{N}{V}$ $[n] = \frac{1}{m^3} = m^{-3}$	$N = \frac{72 \cdot 10^{-3}}{18 \cdot 10^{-3}} \cdot 6 \cdot 10^{23} = 24 \cdot 10^{23}$ $V = \frac{72 \cdot 10^{-3}}{1000} = 72 \cdot 10^{-6}\ m^3$ $n = \frac{24 \cdot 10^{23}}{72 \cdot 10^{-6}} = 3{,}33 \cdot 10^{28}\ m^{-3}$ **Javob:** $n = 3{,}33 \cdot 10^{28}\ m^{-3}$
Topish kerak: n =?		

⑤. Hajmi 6 cm^3 bo‘lgan olmosdagi atomlar sonini aniqlang. Olmosning zichligi 3500 kg/m^3 va molyar massasi 12 g/mol.

Berilgan:	Formulasi:	Hisoblash:
$V = 6\ cm^3$ $\rho = 3500\ kg/m^3$ $M = 12\ g/mol$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$	$m = \rho \cdot V$ $N = \frac{m}{M} \cdot N_A$ $[N] = \frac{g}{g/mol} \cdot \frac{1}{mol} = 1$	$m = 3500 \cdot 6 \cdot 10^{-6} = 21\ g$ $N = \frac{21}{12} \cdot 6 \cdot 10^{23} = 1{,}05 \cdot 10^{24}\ ta$ **Javob:** $N = 1{,}05 \cdot 10^{24}\ ta$
Topish kerak: N =?		

⑥. Modda miqdori 200 mol bo‘lgan misdan qalinligi 2 mm bo‘lgan tekis mis plastinkasi yasalgan. Plastinkaning yuzasi nimaga teng? Misning zichligi 8900 kg/m^3 va molyar massasi 64 g/mol.

Berilgan:	Formulasi:	Hisoblash:
$\nu = 200\ mol$ $h = 2\ mm = 2 \cdot 10^{-3}\ m$ $\rho = 8900\ kg/m^3$ $M = 64 \cdot 10^{-3}\ kg/mol$	$\nu = \frac{m}{M},\ \ m = \nu \cdot M$ $m = \rho \cdot V = \rho \cdot S \cdot h$ $S = \frac{m}{\rho \cdot h} = \frac{\nu \cdot M}{\rho \cdot h}$ $[S] = \frac{kg}{kg/m^3 \cdot m} = m^2$	$S = \frac{200 \cdot 64 \cdot 10^{-3}}{8900 \cdot 2 \cdot 10^{-3}} \approx 0{,}72\ m^2$ **Javob:** $S \approx 0{,}72\ m^2$
Topish kerak: S =?		

⑦. Moddaning zichligi 5 g/cm^3 bo‘lsa, to‘la sirtining yuzi 24 cm^2 bo‘lgan kubning massasi qanday bo‘ladi?

Berilgan:	Formulasi:	Hisoblash:
$\rho = 5\ g/cm^3$ $S_T = 24\ cm^2$	$S_T = 6a^2,\ \ a = \sqrt{\frac{S_T}{6}}$ $m = \rho \cdot V = \rho \cdot a^3$ $[m] = g/cm^3 \cdot cm^3 = g$	$a = \sqrt{\frac{24}{6}} = 2\ cm$ $m = 5 \cdot 2^3 = 40\ g$ **Javob:** $m = 40\ g$
Topish kerak: m =?		

⑧. Modda miqdori 34 mol simob qancha hajmni egallaydi? Simobning zichligi 13,6 g/cm^3, molyar massasi 200 g/mol ga teng deb oling.

Berilgan:	Formulasi:	Hisoblash:
$\nu = 34\ mol$ $\rho = 13{,}6\ g/cm^3$ $M = 200\ g/mol$	$m = \nu \cdot M$ $V = \frac{m}{\rho} = \frac{\nu \cdot M}{\rho}$ $[V] = \frac{mol \cdot g/mol}{g/cm^3} = cm^3$	$V = \frac{34 \cdot 200}{13{,}6} = 500\ cm^3 = 0{,}5\ l$ **Javob:** $V = 0{,}5\ l$
Topish kerak: V =?		

⑨. 10 l hajmli idishga 1,6 kg massali kislorod solingan. Idishdagi gaz molekulalarining konsentratsiyasini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$V = 10\ l = 10^{-2}\ m^3$ $m = 1{,}6\ kg$ $M = 32 \cdot 10^{-3}\ kg/mol$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$	$N = \frac{m}{M} \cdot N_A$ $n = \frac{N}{V}$ $[n] = \frac{1}{m^3} = m^{-3}$	$N = \frac{1{,}6}{32 \cdot 10^{-3}} \cdot 6 \cdot 10^{23} = 3 \cdot 10^{25}$ $n = \frac{3 \cdot 10^{25}}{10^{-2}} = 3 \cdot 10^{27}\ m^{-3}$
Topish kerak: n =?		**Javob:** $n = 3 \cdot 10^{27}\ m^{-3}$

⑩*. Suv molekulasining diametrini $3 \cdot 10^{-10}$ m deb, 3 g suvdagi barcha molekulalar bir-biriga zich qilib bir qatorga joylashtirilsa, qanday uzunlik hosil bo‘lishini hisoblang. Bu uzunlikni Yerdan Oygacha bo‘lgan masofa ($3{,}84 \cdot 10^5$ km) bilan taqqoslang.

Berilgan:	Formulasi:	Hisoblash:
$d = 3 \cdot 10^{-10}\ m$ $m = 3\ g$ $M = 18\ g/mol$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$ $S = 3{,}84 \cdot 10^5\ km$	$N = \frac{m}{M} \cdot N_A$ $l = N \cdot d$ $[l] = 1 \cdot m = m$	$N = \frac{3}{18} \cdot 6 \cdot 10^{23} = 10^{23}\ ta$ $l = 10^{23} \cdot 3 \cdot 10^{-10} = 3 \cdot 10^{13}\ m$ $\frac{l}{S} = \frac{3 \cdot 10^{13}}{3{,}84 \cdot 10^8} = 78125$
Topish kerak: l =? l/S =?		**Javob:** $l = 3 \cdot 10^{13}\ m$ Suv molekulalari bir qator qilib joylash-tirgandagi uzunlik Yerdan Oygacha bo‘lgan masofadan 78125 marta katta

⑪*. Idishdagi suvda $3 \cdot 10^{24}$ ta suv molekulasi bo‘lsa, suvning hajmi qanday? Suv molekulasining diametri $3\ 10^{-10}$ m ga teng.

Berilgan:	Formulasi:	Hisoblash:
$N = 3 \cdot 10^{24}$ $d = 3 \cdot 10^{-10}\ m$	$V = N \cdot d^3$	$V = 3 \cdot 10^{24} \cdot (3 \cdot 10^{-10})^3 = 81 \cdot 10^{-6}\ m^3$
Topish kerak: V =?	$[V] = 1 \cdot m^3 = m^3$	**Javob:** $V = 81 \cdot 10^{-6}\ m^3 = 81\ cm^3$

⑫*. Moy molekulasining diametri taxminan $2 \cdot 10^{-10}$ m ga teng bo‘lsa, 24 cm^3 moyda qancha moy molekulasi borligini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$d = 2 \cdot 10^{-10}\ m$ $V = 24\ cm^3 = 24 \cdot 10^{-6}\ m^3$	$N = \frac{V}{d^3}$	$N = \frac{24 \cdot 10^{-6}}{(2 \cdot 10^{-10})^3} = 3 \cdot 10^{24}\ ta$
Topish kerak: N =?	$[N] = \frac{m^3}{m^3} = 1$	**Javob:** $N = 3 \cdot 10^{24}\ ta$

⑬. Biror modda molekulasining diametri fotosuratda 0,5 mm ga teng. Agar fotosurat elektron mikroskop yordamida 200 000 marta kattalashtirib olingan bo‘lsa, ayni modda molekulasining haqiqiy diametiri qancha?

Berilgan:	Formulasi:	Hisoblash:
$d = 0{,}5\ mm = 5 \cdot 10^{-4}\ m$ $K = 200000$	$d_0 = \frac{d}{K}$	$d_0 = \frac{5 \cdot 10^{-4}}{2 \cdot 10^5} = 2{,}5 \cdot 10^{-9}\ m$
Topish kerak: $d_0 = ?$	$[N] = \frac{m^3}{m^3} = 1$	**Javob:** $d_0 = 2{,}5 \cdot 10^{-9}\ m$

⑭. Nima uchun gulxandan chiqayotgan tutun yuqoriga ko‘tarilgan sari hatto shamol bo‘lmaganda ham ko‘zga ko‘rinmay ketadi?

Javob: Bilamizki yer sharini havo qatlami o‘rab turadi. Gulxandan chiqayotgan tutun havo zarrachalaridan yengil bo‘lganligi sababli yuqoriga ko‘tariladi. Yuqoriga ko‘tarilgan sari Broun harakati natijasida tutun zarralari havo zarrlari bilan to‘xtovsiz va tartibsiz urilishi natijasida atmosferaga sochilib ketadi. Atmosferaga sochilib ketgan zarralarni inson ko‘zining ko‘rishi chegaralanganligi tufayli ko‘rolmaydi.

⑮. Nima sababdan singan chinni yoki sopol idishni yelim bilan yopishtirmasa ularni butun holga keltirib bo‘lmaydi? Axir jism molekulalari orasida tortishish kuchi mavjud-ku!

Javob: Qattiq jism zarralarining o‘zaro ta’sir kuchi suyuqlik yoki gazlarga nisbatan kuchliroq. Singan qattiq jismlarni birlashtirish uchun atom o‘lchamlari darajasidagi masofada yaqinlashtirish kerak. Oddiy jarayon bilan bunday masofaga yaqinlashtirishning iloji yo‘q.

Kuchli ta’sirlashuv kichik masofalarda (10^{-15} m) namoyon bo‘ladi. Uning sharofati bilan turli xil yadrolar, atomlar va kimyoviy elementlar mavjud. Kimyoviy elementlarning mavjudligi esa turli murakkab molekulalar, organik birikmalar va hayotning mavjudligini ta’minlovchi omildir.

4-MASHQ

①. Idish vodorod gazi bilan to‘ldirilgan. Idishdagi gaz molekulalarining konsentratsiyasi $4{,}5 \cdot 10^{24}$ m^{-3} ga teng. Idishdagi gaz bosimini hisoblang. Gaz molekulalarining o‘rtacha kvadratik tezligini 400 m/s ga teng deb oling.

Berilgan:	Formulasi:	Hisoblash:
$n = 4{,}5 \cdot 10^{24}\ m^{-3}$ $\bar{\vartheta} = 400\ m/s$ $M = 2 \cdot 10^{-3}\ kg/mol$ $N_A = 6 \cdot 10^{23}\ mol^{-1}$	$m_0 = \frac{M}{N_A}$ $p = \frac{1}{3} n m_0 \overline{\vartheta^2}$	$m_0 = \frac{2 \cdot 10^{-3}}{6 \cdot 10^{23}} \approx 3{,}3 \cdot 10^{-27}\ kg$ $p = \frac{1}{3} \cdot 4{,}5 \cdot 10^{24} \cdot 3{,}3 \cdot 10^{-27} \cdot$
Topish kerak: $p = ?$	$[p] = \frac{1}{m^3} \cdot kg \cdot \frac{m^2}{s^2} = Pa$	$\cdot 400^2 = 792\ Pa$ **Javob:** $p = 792\ Pa$

②. Ideal gaz molekulalarining o‘rtacha kvadratik tezligi 600 m/s va zichligi 0,9 kg/m^3 bo‘lsa, idishdagi gaz bosimini aniqlang.

Berilgan:	**Formulasi:**	**Hisoblash:**
$\bar{\vartheta} = 600\ m/s$ $\rho = 0{,}9\ kg/m^3$	$p = \frac{1}{3}\rho\overline{\vartheta^2}$ $[p] = \frac{kg}{m^3} \cdot \frac{m^2}{s^2} = Pa$	$p = \frac{1}{3} \cdot 0{,}9 \cdot 600^2 = 108000\ Pa = 108\ kP$ **Javob:** $p = 108\ kPa$
Topish kerak: p =?		

③. Idishdagi gazning zichligi 1,5 kg/m^3 va bosimi 7,2 kPa bo‘lgan gaz molekulalarining o‘rtacha kvadratik tezligi nimaga teng?

Berilgan:	**Formulasi:**	**Hisoblash:**
$p = 7{,}2 \cdot 10^3\ Pa$ $\rho = 1{,}5\ kg/m^3$	$\bar{\vartheta} = \sqrt{\frac{3P}{\rho}}$ $[\bar{\vartheta}] = \sqrt{\frac{Pa}{kg/m^3}} = \sqrt{\frac{N/m^2}{kg/m^3}} =$ $= \sqrt{\frac{kg \cdot m/s^2}{kg/m}} = m/s$	$\bar{\vartheta} = \sqrt{\frac{3 \cdot 7{,}2 \cdot 10^3}{1{,}5}} = 120\ m/s$ **Javob:** $\bar{\vartheta} = 120\ m/s$
Topish kerak: $\bar{\vartheta}$ =?		

④. Idishdagi hajm birligidagi molekulalar soni $3 \cdot 10^{25}$ m^{-3} va bosimi 80 kPa bo‘lgan gaz molekulalarining o‘rtacha kinetik energiyasini hisoblang.

Berilgan:	**Formulasi:**	**Hisoblash:**
$n = 3 \cdot 10^{25}\ m^{-3}$ $p = 80 \cdot 10^3\ Pa$	$p = \frac{2}{3}n\bar{E}_k \Rightarrow \bar{E}_k = \frac{3p}{2n}$ $[\bar{E}_k] = \frac{Pa}{m^{-3}} = \frac{N/m^2}{m^{-3}} = N \cdot m = J$	$\bar{E}_k = \frac{3 \cdot 80 \cdot 10^3}{2 \cdot 3 \cdot 10^{25}} = 4 \cdot 10^{-21}\ J$ **Javob:** $\bar{E}_k = 4 \cdot 10^{-21}\ J$
Topish kerak: $\bar{E}_k$ =?		

⑤. Idishga solingan kislorod gazi, idish devoriga 90 kPa bosim bermoqda. Agar kislorod molekulalari 600 m/s o‘rtacha kvadratik tezlik bilan harakatlanayotgan bo‘lsa, idishdagi gaz zichligi qanday bo‘lgan?

Berilgan:	**Formulasi:**	**Hisoblash:**
$p = 90 \cdot 10^3\ Pa$ $\bar{\vartheta} = 600\ m/s$	$\rho = \frac{3P}{\overline{\vartheta^2}}$ $[\rho] = \frac{Pa}{m^2/s^2} = \frac{N/m^2}{m^2/s^2} =$ $= \frac{kg \cdot m \cdot s^2}{m^4 \cdot s^2} = \frac{kg}{m^3}$	$\rho = \frac{3 \cdot 90 \cdot 10^3}{600^2} = 0{,}75\ kg/m^3$ **Javob:** $\rho = 0{,}75\ kg/m^3$
Topish kerak: ρ =?		

⑥. Massasi 0,3 kg bo‘lgan gaz 400 kPa bosimda 1 m^3 hajmni egallasa, uning molekulalari harakatining o‘rtacha kvadratik tezliklari qanday bo‘ladi?

Berilgan:	Formulasi:	Hisoblash:
$m = 0{,}3\ kg$ $p = 400 \cdot 10^3\ Pa$ $V = 1\ m^3$	$\bar{\vartheta} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3PV}{m}}$	$\bar{\vartheta} = \sqrt{\frac{3 \cdot 400 \cdot 10^3 \cdot 1}{0{,}3}} = 200\ m/s$
Topish kerak: $\bar{\vartheta} = ?$		**Javob:** $\bar{\vartheta} = 2000\ m/s$
	$[\bar{\vartheta}] = \sqrt{\frac{Pa}{kg/m^3}} = \sqrt{\frac{N/m^2}{kg/m^3}} = \sqrt{\frac{kg \cdot m/s^2}{kg/m}} = m/s$	

⑦. 30 kPa bosimda bir atomli gaz molekulasining o‘rtacha kinetik energiyasini toping. Berilgan bosimda bu gaz molekulalarining konsentratsiyasi $4 \cdot 10^{25}\ m^{-3}$ ga teng.

Berilgan:	Formulasi:	Hisoblash:
$p = 30 \cdot 10^3\ Pa$ $n = 4 \cdot 10^{25}\ m^{-3}$	$p = \frac{2}{3} n\bar{E}_k \Rightarrow \bar{E}_k = \frac{3p}{2n}$	$\bar{E}_k = \frac{3 \cdot 30 \cdot 10^3}{2 \cdot 4 \cdot 10^{25}} = 1{,}125 \cdot 10^{-21}\ J$
Topish kerak: $\bar{E}_k = ?$	$[\bar{E}_k] = \frac{Pa}{m^{-3}} = \frac{N/m^2}{m^{-3}} = N \cdot m = J$	**Javob:** $\bar{E}_k = 1{,}125 \cdot 10^{-21}\ J$

5-MASHQ

①. Selsiy shkalasida ifodalangan temperaturalarni Kelvin shkalasida ifodalang:
0 °C, 27 °C, 100 °C, 127 °C, -73 °C, -223 °C, -200 °C.

Berilgan:	Formulasi:	Hisoblash:
$t_1 = 0\ °C,\ t_2 = 27\ °C,$ $t_3 = 100\ °C,\ t_4 = 127\ °C,$ $t_5 = -73\ °C,\ t_6 = -223\ °C,$ $t_7 = -200\ °C$	$T = t + 273$	$T_1 = 0 + 273 = 273\ K$ ….. $T_7 = -200 + 273 = 73\ K$
Topish kerak: $T_{1,2,3,4,5,6,7} = ?$	**Javob:** 273 K, 300 K 373 K, 400 K, 200 K, 50 K, 73 K	

②. Quyidagi Kelvin shkalalarida ifodalangan temperaturalarni Selsiy shkalalarida ifodalang: 0 K, 73 K, 273 K, 300 K, 773 K, 1000 K, 2000 K.

Berilgan:	Formulasi:	Hisoblash:
$T_1 = 0\ K,\ T_2 = 73\ K$ $T_3 = 273\ K,\ T_4 = 300\ K$ $T_5 = 773\ K,\ T_6 = 1000\ K$ $T_7 = 2000\ K$	$t = T - 273$	$t_1 = 0 - 273 = -273\ °C$ …. $t_7 = 2000 - 273 = 1727\ °C$
Topish kerak: $t_{1,2,3,4,5,6,7} = ?$		**Javob:** -273 °C , -200 °C, 0 °C, 27 °C, 500 °C, 727 °C, 1727 °C

③. Yopiq idishdagi gaz 27 °C dan 627 °C gacha qizdirildi. Bunda gaz molekulalarining idish devoriga beradigan bosimi qanday o‘zgaradi?

Berilgan:	Formulasi:	Hisoblash:
$t_1 = 27\ °C \Rightarrow\ T_1 = 300\ K$ $t_2 = 627\ °C \Rightarrow\ T_2 = 900\ K$	$p = nkT$ $\frac{p_2}{p_1} = \frac{nkT_2}{nkT_1} = \frac{T_2}{T_1}$	$\frac{p_2}{p_1} = \frac{900}{300} = 3$
Topish kerak: $p_2/p_1 = ?$		**Javob:** 3 *marta ortadi*

④. Idishdagi gaz molekulalarining konsentratsiyasi $3 \cdot 10^{27}$ m^{-3}ga teng. Idish ichidagi temperatura 60 °C bo‘lganda gaz molekulalarining idish devoriga beradigan bosimi qanday bo‘ladi?

Berilgan:	Formulasi:	Hisoblash:
$n = 3 \cdot 10^{27}\ m^{-3}$ $t = 60\ °C \Rightarrow\ T = 333\ K$ $k = 1{,}38 \cdot 10^{-23}\ J/K$	$p = nkT$ $[p] = \frac{1}{m^3} \cdot \frac{J}{K} \cdot K = \frac{J}{m^3}$ $= \frac{N \cdot m}{m^3} = Pa$	$p = 3 \cdot 10^{27} \cdot 1{,}38 \cdot 10^{-23} \cdot 333 =$ $\approx 13{,}8 \cdot 10^6\ Pa = 13{,}8\ MPa$ **Javob:** $p \approx 13{,}8\ MPa$
Topish kerak: $p = ?$		

⑤. Idish ichidagi gazning temperaturasi 400 K bo‘lganda, manometr idishdagi gaz bosimi 276 kPa ga teng bo‘lganligini ko‘rsatdi. Idishdagi gaz molekulalarining konsentratsiyasi nimaga teng?

Berilgan:	Formulasi:	Hisoblash:
$p = 276 \cdot 10^3\ Pa$ $T = 400\ K$ $k = 1{,}38 \cdot 10^{-23}\ \frac{J}{K}$	$p = nkT \Rightarrow\ n = \frac{p}{kT}$ $[n] = \frac{Pa}{J/K \cdot K} = \frac{N/m^2}{N \cdot m} = m^{-3}$	$n = \frac{276 \cdot 10^3}{1{,}38 \cdot 10^{-23} \cdot 400} =$ $= 13{,}8 \cdot 10^6\ Pa = 5 \cdot 10^{25}\ m^{-3}$ **Javob:** $n = 5 \cdot 10^{25}\ m^{-3}$
Topish kerak: $n = ?$		

⑥. Normal sharoitda 1 m^3 hajmdagi havo molekulalarining sonini baholang. Normal sharoit uchun bosimni 100 kPa, temperaturani 273 K ga teng deb oling.

Berilgan:	Formulasi:	Hisoblash:
$V = 1\ m^3$ $p = 100 \cdot 10^3\ Pa$ $T = 273\ K$ $k = 1{,}38 \cdot 10^{-23} J/K$	$n = \frac{p}{kT}$ $N = n \cdot V = \frac{pV}{kT}$ $[N] = \frac{1}{m^3} \cdot m^3 = 1$	$N = \frac{100 \cdot 10^3 \cdot 1}{1{,}38 \cdot 10^{-23} \cdot 273} \approx 2{,}67 \cdot 10^{25}\ ta$ **Javob:** $N \approx 2{,}65 \cdot 10^{25}\ ta$
Topish kerak: $N = ?$		

⑦. Maxsus so‘ruvchi nasos yordamida idishdan havo so‘rilib, uning ichida 1 pPa bosimdagi vakuum hosil qilindi. Vakuumning 1 cm^3 hajmida qancha gaz molekulasi bor? Idish ichidagi temperatura 300 K.

Berilgan:	Formulasi:	Hisoblash:
$p = 1\,pPa = 10^{-12}\,Pa$ $V = 1\,cm^3 = 10^{-6}\,m^3$ $T = 300\,K$ $k = 1{,}38 \cdot 10^{-23} J/K$	$n = \frac{p}{kT}$ $N = n \cdot V = \frac{pV}{kT}$ $[N] = \frac{1}{m^3} \cdot m^3 = 1$	$N = \frac{10^{-12} \cdot 10^{-6}}{1{,}38 \cdot 10^{-23} \cdot 300} \approx 242\ ta$ **Javob:** $N \approx 242\ ta$
Topish kerak: $N = ?$		

6-MASHQ

①. Vodorod molekulasining -23 °C temperaturadagi o‘rtacha kvadratik tezligini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$t = -23\,°C, T = 250\,K$ $M = 2 \cdot 10^{-3}\,kg/mol$ $R = 8{,}31\,J/mol \cdot K$	$\bar{\vartheta} = \sqrt{\frac{3RT}{M}}$	$\bar{\vartheta} = \sqrt{\frac{3 \cdot 8{,}31 \cdot 250}{2 \cdot 10^{-3}}} \approx 1765\,m/s$ **Javob:** $\bar{\vartheta} \approx 1934\,m/s$
Topish kerak: $\bar{\vartheta} = ?$	$[\bar{\vartheta}] = \sqrt{\frac{(J/mol \cdot K) \cdot K}{kg/mol}} = \sqrt{\frac{J}{kg}} = \sqrt{\frac{kg \cdot m^2/s^2}{kg}} = \frac{m}{s}$	

②. Qanday temperaturada kislorod molekulasining o‘rtacha kvadratik tezligi 500 m/s ga teng bo‘ladi?

Berilgan:	Formulasi:	Hisoblash:
$\bar{\vartheta} = 500\,m/s$ $M = 32 \cdot 10^{-3}\,kg/mol$ $R = 8{,}31\,J/mol \cdot K$	$T = \frac{\overline{\vartheta^2} M}{3R}$	$T = \frac{500^2 \cdot 32 \cdot 10^{-3}}{3 \cdot 8{,}31} \approx 321\,K$ **Javob:** $T \approx 321\,K$
Topish kerak: $T = ?$	$[T] = \frac{(m^2/s^2) \cdot kg/mol}{J/mol \cdot K} = \frac{(m^2/s^2) \cdot kg \cdot K}{kg \cdot m^2/s^2} = K$	

③. Gaz molekulalari ilgarilanma harakatining o‘rtacha kinetik energiyasi $9{,}52 \cdot 10^{-21}$ J bo‘lgan gazning temperaturasini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$\bar{E}_k = 9{,}52 \cdot 10^{-21}\,J$ $k = 1{,}38 \cdot 10^{-23} J/K$	$\bar{E}_k = \frac{3}{2} \cdot kT, \quad T = \frac{2\bar{E}_k}{3k}$ $[\bar{E}_k] = J/(J/K) = K$	$T = \frac{2 \cdot 9{,}52 \cdot 10^{-21}}{3 \cdot 1{,}38 \cdot 10^{-23}} \approx 460\,K$ **Javob:** $T \approx 460\,K$
Topish kerak: $T = ?$		

④. Molekulalar konsentratsiyasi $4 \cdot 10^{26}$ m^{-3} va bosimi $1{,}6 \cdot 10^5$ Pa bo‘lgan bir atomli gaz molekulalarining o‘rtacha kinetik energiyasi nimaga teng?

Berilgan:	**Formulasi:**	**Hisoblash:**
$n = 4 \cdot 10^{26}\ m^{-3}$ $p = 1{,}6 \cdot 10^5\ Pa$ $k = 1{,}38 \cdot 10^{-23} J/K$	$p = \frac{2}{3} \cdot n\bar{E}_k,\ \ \bar{E}_k = \frac{3p}{2n}$	$\bar{E}_k = \frac{3 \cdot 1{,}6 \cdot 10^5}{2 \cdot 4 \cdot 10^{26}} = 6 \cdot 10^5\ J$
Topish kerak: $\bar{E}_k$ =?	$[\bar{E}_k] = \frac{Pa}{m^{-3}} = \frac{N/m^2}{1/m^3} = J$	**Javob:** $\bar{E}_k = 6 \cdot 10^5\ J$

⑤. Bir atomli gaz molekulalarining ilgarilanma harakati o‘rtacha kinetik energiyasi $1{,}2 \cdot 10^{-20}$ J va bosimi 2,4 MPa bo‘lsa, shu gaz molekulalarining konsentratsiyasi qanday bo‘ladi?

Berilgan:	**Formulasi:**	**Hisoblash:**
$\bar{E}_k = 1{,}2 \cdot 10^{-20}\ J$ $p = 2{,}4 \cdot 10^6\ Pa$	$p = \frac{2}{3} \cdot n\bar{E}_k,\ \ n = \frac{3p}{2\bar{E}_k}$	$n = \frac{3 \cdot 2{,}4 \cdot 10^6}{2 \cdot 1{,}2 \cdot 10^{-20}} = 3 \cdot 10^{26}\ m^{-3}$
Topish kerak: n =?	$[n] = \frac{Pa}{J} = \frac{\frac{N}{m^2}}{N \cdot m} = m^{-3}$	**Javob:** $n = 3 \cdot 10^{26}\ m^{-3}$

⑥. Miqdori ikki mol bo‘lgan gazning idish devorlariga beradigan bosimi 10 kPa ga teng. Gaz egallab turgan hajmni aniqlang. Gazning temperaturasi 300 K.

Berilgan:	**Formulasi:**	**Hisoblash:**
$\nu = 2\ mol$ $p = 10^4\ Pa$ $T = 300\ K$ $R = 8{,}31\ J/mol \cdot K$	$p = nkT,\ \ n = \frac{p}{kT}$ $N = \nu N_A,\ \ R = k \cdot N_A$ $V = \frac{N}{n} = \frac{\nu k N_A T}{p} = \frac{\nu RT}{p}$	$V = \frac{2 \cdot 8{,}31 \cdot 300}{10^4} \approx 0{,}5\ m^3$ **Javob:** $V = 0{,}5\ m^3$
Topish kerak: V =?	$[V] = \frac{mol \cdot (J/mol \cdot K) \cdot K}{Pa} = \frac{J}{Pa} = \frac{N \cdot m}{N/m^2} = m^3$	

⑦*. Qanday temperaturadagi geliy molekulalarining o‘rtacha kvadratik tezligi, 350 K temperaturadagi vodorod molekulalarining o‘rtacha kvadratik tezligiga teng bo‘ladi?

Berilgan:	**Formulasi:**	**Hisoblash:**
$\bar{\vartheta}_1 = \bar{\vartheta}_2$ $T_1 = 350\ K$ $M_1 = 2 \cdot 10^{-3}\ kg/mol$ $M_2 = 4 \cdot 10^{-3}\ kg/mol$	$\bar{\vartheta} = \sqrt{3RT/M},\ \ \bar{\vartheta}_1 = \bar{\vartheta}_2$ $\frac{3RT_1}{M_2} = \frac{3RT_1}{M_2},\ \ \frac{T_1}{M_2} = \frac{T_2}{M_2}$ $T_2 = \frac{T_1 \cdot M_2}{M_2}$	$T_2 = \frac{350 \cdot 4 \cdot 10^{-3}}{2 \cdot 10^{-3}} = 700\ K$ **Javob:** $T = 700\ K$
Topish kerak: T_2 =?	$[T] = \frac{K \cdot kg/mol}{kg/mol} = K$	

⑧*. Gaz temperaturasi 150°C ga oshirilganda, molekulalarining o‘rtacha kvadratik tezligi 300 m/s dan 600 m/s gacha ortdi. Gazning dastlabki temperaturasi qanday bo‘lgan?

Berilgan:	Formulasi:	Hisoblash:
$T_2 = T_1 + \Delta T$ $\Delta T = 150\ K$ $\bar{\vartheta}_1 = 300\ m/s$ $\bar{\vartheta}_2 = 600\ m/s$ *Topish kerak:* $T_1 = ?$	$\bar{\vartheta}_1 = \sqrt{\frac{3RT_1}{M}}$ $\bar{\vartheta}_2 = \sqrt{\frac{3RT_2}{M}} = \sqrt{\frac{3R(T_1+\Delta T)}{M}}$ $\frac{\bar{\vartheta}_2}{\bar{\vartheta}_1} = \sqrt{\frac{T_1+\Delta T}{T_1}},\ \left(\frac{\bar{\vartheta}_2}{\bar{\vartheta}_1}\right)^2 = \frac{T_1+\Delta T}{T_1}$ $T_1 = \frac{\Delta T}{\left(\frac{\bar{\vartheta}_2}{\bar{\vartheta}_1}\right)^2 - 1}$ $[T] = \frac{K}{\left(\frac{m/s}{m/s}\right)^2} = K$	$T_1 = \frac{150}{\left(\frac{600}{300}\right)^2 - 1} = 50\ K$ **Javob:** $T = 50\ K$

7-MASHQ

①. Bosimi 0,45 MPa va temperaturasi 52°C bo‘lganda 500 mol gaz qanday hajmni egallaydi?

Berilgan:	Formulasi:	Hisoblash:
$p = 0{,}45 \cdot 10^6\ Pa$ $T = 325\ K$ $\nu = 500\ mol$ $R = 8{,}31\ J/mol \cdot K$ *Topish kerak:* $V = ?$	$pV = \nu RT,\quad V = \frac{\nu RT}{p}$ $[V] = \frac{mol \cdot (J/mol \cdot K) \cdot K}{Pa} = \frac{J}{Pa} = \frac{N \cdot m}{N/m^2} = m^3$	$V = \frac{500 \cdot 8{,}31 \cdot 325}{0{,}45 \cdot 10^6} \approx 3\ m^3$ **Javob:** $V = 3\ m^3$

②. Hajmi 0,05 m^3 va temperaturasi 500 K bo‘lgan gazning bosimi 250 kPa. Modda miqdorini aniqlang.

Berilgan:	Formulasi:	Hisoblash:
$V = 0{,}05\ m^3$ $T = 500\ K$ $p = 250 \cdot 10^3\ Pa$ $R = 8{,}31\ J/mol \cdot K$ *Topish kerak:* $\nu = ?$	$pV = \nu RT,\quad \nu = \frac{pV}{RT}$ $[\nu] = \frac{Pa \cdot m^3}{(J/mol \cdot K) \cdot K} = \frac{N/m^2 \cdot m^3 \cdot mol}{N \cdot m} = mol$	$\nu = \frac{250 \cdot 10^3 \cdot 0{,}05}{8{,}31 \cdot 500} = 3\ mol$ **Javob:** $\nu = 3\ mol$

③. Massasi 8 g bo‘lgan gaz 27 °C temperaturada va 150 kPa bosimda 4,15 *l* hajmni egallaydi. Bu qanday gaz?

Berilgan:	**Formulasi:**	**Hisoblash:**
$m = 8\ g = 8 \cdot 10^{-3}\ kg$ $T = 300\ K$ $p = 150 \cdot 10^3\ Pa$ $V = 4{,}15\ l = 4{,}15 \cdot 10^{-3}\ m^3$ $R = 8{,}31\ J/mol \cdot K$	$pV = \frac{m}{M}RT$ $M = \frac{mRT}{pV}$	$M = \frac{8 \cdot 10^{-3} \cdot 8{,}31 \cdot 300}{150 \cdot 10^3 \cdot 4{,}15 \cdot 10^{-3}} =$ $= 32 \cdot 10^{-3} kg/mol = 32 g/mol$ **Javob:** $M = 32\ g/mol$ (kislorod gazi O_2)
Topish kerak: $\nu = ?$	$[\nu] = \frac{kg \cdot \frac{J}{mol \cdot K} \cdot K}{Pa \cdot m^3} = \frac{kg \cdot N \cdot m}{\frac{N}{m^2} \cdot m^3 \cdot mol} = kg/mol$	

④. Temperaturasi 367 °C va bosimi 415 kPa bo‘lgan kislorod gazining zichligi qanday?

Berilgan:	**Formulasi:**	**Hisoblash:**
$T = 640\ K$ $p = 415 \cdot 10^3\ Pa$ $M = 32\ g/mol$ $R = 8{,}31\ J/mol \cdot K$	$pV = \frac{m}{M}RT,\ \ m = \rho \cdot V$ $pV = \frac{\rho V}{M}RT,\ \ \rho = \frac{pM}{RT}$	$\rho = \frac{415 \cdot 10^3 \cdot 32 \cdot 10^{-3}}{8{,}31 \cdot 640} =$ $= 2{,}5\ kg/m^3$ **Javob:** $\rho = 2{,}5\ kg/m^3$
Topish kerak: $\rho = ?$	$[\rho] = \frac{Pa \cdot kg/mol}{(J/mol \cdot K) \cdot K} = \frac{N/m^2 \cdot kg}{N \cdot m} = kg/m^3$	

⑤. 24 *l* hajmli ballonda 1,2 kg karbonat angidrid gazi bor. Ballon $3 \cdot 10^6$ Pa bosimgacha chidaydi. Qanday temperaturada portlash xavfi tug‘iladi?

Berilgan:	**Formulasi:**	**Hisoblash:**
$V = 24\ l = 24 \cdot 10^{-3}\ m^3$ $m = 1{,}2\ kg$ $p = 3 \cdot 10^6\ Pa$ $M = 44 \cdot 10^{-3}\ kg/mol$ $R = 8{,}31\ J/mol \cdot K$	$pV = \frac{m}{M}RT$ $T = \frac{pVM}{mR}$	$T = \frac{3 \cdot 10^6 \cdot 24 \cdot 10^{-3} \cdot 44 \cdot 10^{-3}}{1{,}2 \cdot 8{,}31} = 3\ K$ **Javob:** $T \approx 318\ K$
Topish kerak: $T = ?$	$[T] = \frac{Pa \cdot m^3 \cdot kg/mol}{kg \cdot (J/mol \cdot K)} = \frac{N/m^2 \cdot m^3 \cdot K}{N \cdot m} = K$	

⑥. Hajmi 40 *l* bo‘lgan idishga gaz solingan bo‘lib, uning temperaturasi 400 K va bosimi 200 kPa ga teng. Idishdagi gazning miqdorini aniqlang.

Berilgan:	**Formulasi:**	**Hisoblash:**
$V = 40\ l = 40 \cdot 10^{-3}\ m^3$ $T = 400\ K$ $p = 2 \cdot 10^5\ Pa$ $R = 8{,}31\ J/mol \cdot K$	$pV = \nu RT$ $\nu = \frac{pV}{RT}$	$\nu = \frac{2 \cdot 10^5 \cdot 40 \cdot 10^{-3}}{8{,}31 \cdot 400} = 2{,}4\ mol$ **Javob:** $\nu = 2{,}4\ mol$
Topish kerak: $\nu = ?$	$[\nu] = \frac{Pa \cdot m^3}{(J/mol \cdot K) \cdot K} = \frac{N/m^2 \cdot m^3 \cdot mol}{N \cdot m} = mol$	

⑦. Temperaturasi 17 °C bo‘lgan 4x5x3 m^3 o‘lchamli xonadagi havo miqdorini aniqlang. Atmosfera bosimi 10^5 Pa ga teng deb oling.

Berilgan:	**Formulasi:**	**Hisoblash:**
$T = 290\ K$ $V = 60\ m^3$ $p = 10^5\ Pa$ $R = 8{,}31\ J/mol \cdot K$	$pV = \nu RT$ $\nu = \frac{pV}{RT}$	$\nu = \frac{10^5 \cdot 60}{8{,}31 \cdot 290} \approx 2490\ mol$ **Javob:** $\nu \approx 2490\ mol$
Topish kerak: $\nu = ?$	$[\nu] = \frac{Pa \cdot m^3}{(J/mol \cdot K) \cdot K} = \frac{N/m^2 \cdot m^3 \cdot mol}{N \cdot m} = mol$	

⑧. Hajm 16,6 *l* bo‘lgan idishda 280 g azot gazi 3,5 MPa bosim ostida bo‘lsa, uning temperaturasi nimaga teng?

Berilgan:	**Formulasi:**	**Hisoblash:**
$V = 16{,}6\ l = 16{,}6 \cdot 10^{-3}\ m^3$ $m = 280\ g = 0{,}28\ kg$ $p = 3{,}5\ MPa = 3{,}5 \cdot 10^6\ Pa$ $M = 28 \cdot 10^{-3}\ kg/mol$ $R = 8{,}31\ J/mol \cdot K$	$pV = \frac{m}{M}RT$ $T = \frac{pVM}{mR}$	$T = \frac{3{,}5 \cdot 10^6 \cdot 16{,}6 \cdot 10^{-3} \cdot 28 \cdot 10^{-3}}{0{,}28 \cdot 8{,}31} = 700\ K$ **Javob:** $T = 700\ K$
Topish kerak: $T = ?$	$[T] = \frac{Pa \cdot m^3 \cdot kg/mol}{kg \cdot (J/mol \cdot K)} = \frac{N/m^2 \cdot m^3 \cdot K}{N \cdot m} = K$	